1. Write to a File

Creates a file and writes content into it.

```
#include <stdio.h>

int main() {
    FILE *fp;
    fp = fopen("file.txt", "w");

    fprintf(fp, "Hello, this is file writing example.\n");

    fclose(fp);
    return 0;
}
```

---

2. Read from a File

Reads content from an existing file.

```
#include <stdio.h>

int main() {
    FILE *fp;
    char ch;

    fp = fopen("file.txt", "r");

    while ((ch = fgetc(fp)) != EOF) {
        printf("%c", ch);
    }

    fclose(fp);
    return 0;
}
```

---

3. Append to a File

Adds content at the end of the file.

```
#include <stdio.h>

int main() {
    FILE *fp;

    fp = fopen("file.txt", "a");

    fprintf(fp, "This line is appended.\n");
```

```
    fclose(fp);
    return 0;
}
```

---

4. Copy Content from One File to Another

```
#include <stdio.h>

int main() {
    FILE *fp1, *fp2;
    char ch;

    fp1 = fopen("file.txt", "r");
    fp2 = fopen("copy.txt", "w");

    while ((ch = fgetc(fp1)) != EOF) {
        fputc(ch, fp2);
    }

    fclose(fp1);
    fclose(fp2);
    return 0;
}
```

---

5. Count Characters in a File

```
#include <stdio.h>

int main() {
    FILE *fp;
    char ch;
    int count = 0;

    fp = fopen("file.txt", "r");

    while ((ch = fgetc(fp)) != EOF) {
        count++;
    }

    printf("Total characters: %d", count);

    fclose(fp);
    return 0;
}
```

---

6. Count Lines in a File

```
#include <stdio.h>

int main() {
```

```
    FILE *fp;
    char ch;
    int lines = 0;

    fp = fopen("file.txt", "r");

    while ((ch = fgetc(fp)) != EOF) {
        if (ch == '\n') {
            lines++;
        }
    }

    printf("Total lines: %d", lines);

    fclose(fp);
    return 0;
}
```

---

7. Read and Write Using `fgets` and `fputs`

```
#include <stdio.h>

int main() {
    FILE *fp;
    char str[100];

    fp = fopen("file.txt", "w");

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    fputs(str, fp);

    fclose(fp);

    fp = fopen("file.txt", "r");

    fgets(str, sizeof(str), fp);
    printf("File content: %s", str);

    fclose(fp);
    return 0;
}
```

Function for formatted input and output.

1. `cin` (Input)

```
#include <iostream>
using namespace std;

int main() {
    int x;
    cin >> x;
    cout << x;
}
```

Input:

```
5
```

Output:

```
5
```

---

2. `cout` (Output)

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello";
}
```

Output:

```
Hello
```

---

3. `setw()`

```
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    cout << setw(5) << 10;
}
```

Output:

```
   10
```

---

4. setprecision()

```
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    double x = 3.14159;
    cout << setprecision(3) << x;
}
```

Output:

```
3.14
```

---

5. fixed

```
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    double x = 3.14159;
    cout << fixed << x;
}
```

Output:

```
3.141590
```

---

6. setfill()

```
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    cout << setw(5) << setfill('*') << 10;
}
```

Output:

```
***10
```

---

7. left

```
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    cout << left << setw(5) << 10;
}
```

Output:

```
10
```

---

8. `right`

```
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    cout << right << setw(5) << 10;
}
```

Output:

```
   10
```

---

9. `endl`

```
#include <iostream>
using namespace std;

int main() {
    cout << "A" << endl << "B";
}
```

Output:

```
A
B
```

---

10. `showpoint`

```
#include <iostream>
using namespace std;

int main() {
    cout << showpoint << 5.0;
}
```

Output:

```
5.00000
```

Quick Combined Example:

```
cout << fixed << setprecision(2) << setw(8) << setfill('*') << 3.5;
```

Output:

```
****3.50
```

1. Write data to a file (`ofstream`)

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream fout("file1.txt");
    fout << "Hello C++ File Handling";
    fout.close();
    cout << "Data written";
}
```

Output:

```
Data written
```

File Content (`file1.txt`):

```
Hello C++ File Handling
```

---

2. Read data from a file (`ifstream`)

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream fin("file1.txt");
    string data;
    getline(fin, data);
    cout << data;
    fin.close();
}
```

Output:

```
Hello C++ File Handling
```

---

3. Append data to a file (`ofstream` with `ios::app`)

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream fout("file1.txt", ios::app);
    fout << "\nAppended Line";
    fout.close();
    cout << "Data appended";
}
```

Output:

```
Data appended
```

Updated File:

```
Hello C++ File Handling
Appended Line
```

---

4. Read entire file using loop

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream fin("file1.txt");
    string line;

    while(getline(fin, line)) {
        cout << line << endl;
    }
    fin.close();
}
```

Output:

```
Hello C++ File Handling
Appended Line
```

---

5. Count number of words in file

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream fin("file1.txt");
```

```
    string word;
    int count = 0;

    while(fin >> word) {
        count++;
    }
    cout << "Words: " << count;
    fin.close();
}
```

Output:

```
Words: 4
```

---

6. Copy one file to another (`ifstream` + `ofstream`)

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream fin("file1.txt");
    ofstream fout("file2.txt");
    string line;

    while(getline(fin, line)) {
        fout << line << endl;
    }

    fin.close();
    fout.close();
    cout << "File copied";
}
```

Output:

```
File copied
```

---

7. Use `fstream` for read + write

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    fstream file("file3.txt", ios::in | ios::out | ios::trunc);

    file << "C++ fstream Example";

    file.seekg(0); // move pointer to beginning
```

```
    string data;
    getline(file, data);
    cout << data;

    file.close();
}
```

Output:

```
C++ fstream Example
```

Quick Viva Tip:

- `ofstream` → write
- `ifstream` → read
- `fstream` → read + write